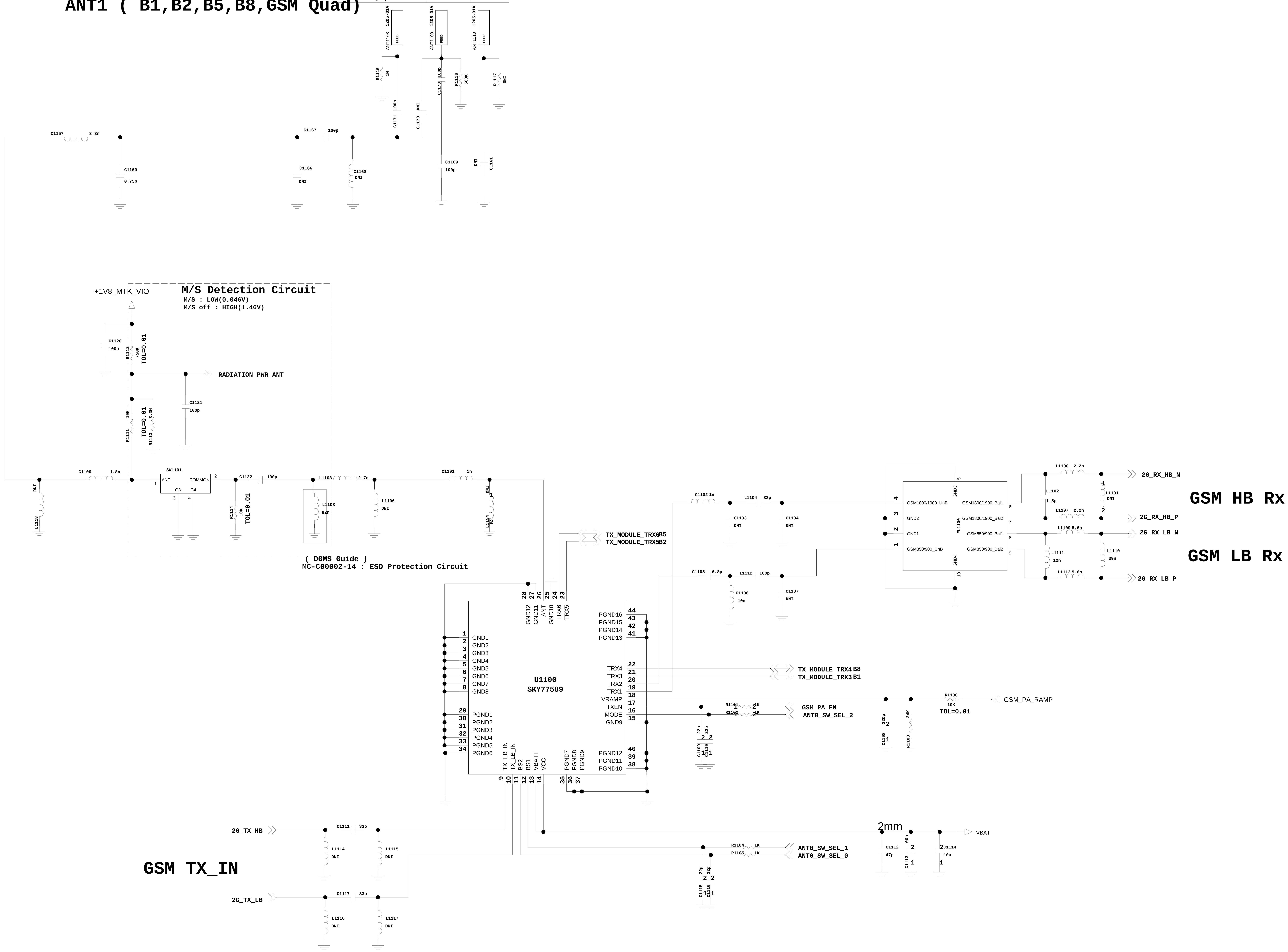
< 1-4-1-3_ANT_WCDMA_Quad>

ANT1_PAD

ANT1 (B1,B2,B5,B8,GSM Quad)

Ref. #	ANT1108	ANT1109	ANT1110
Default	Signal	GND	GND
Option (C1171=NC, C1170=100pF)	NC	Signal	GND



		TXEN	Mode	BS1	BS2
		GSM_PA_EN	ANT_SEL2	ANT_SEL1	ANT_SEL0
Stanby	Stanby	0	0	0	0
LB_TX	G_LB_TX	1	0	0	1
HB_TX	G_HB_TX	1	0	1	1
TRX1	G_HB_RX	0	1	0	0
TRX2	G_LB_RX	0	1	1	0
TRX3	B1	0	1	0	1
TRX4	B8	0	1	1	1
TRX5	B2	0	0	1	0
TRX6	B5	0	0	0	1

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Schematic1

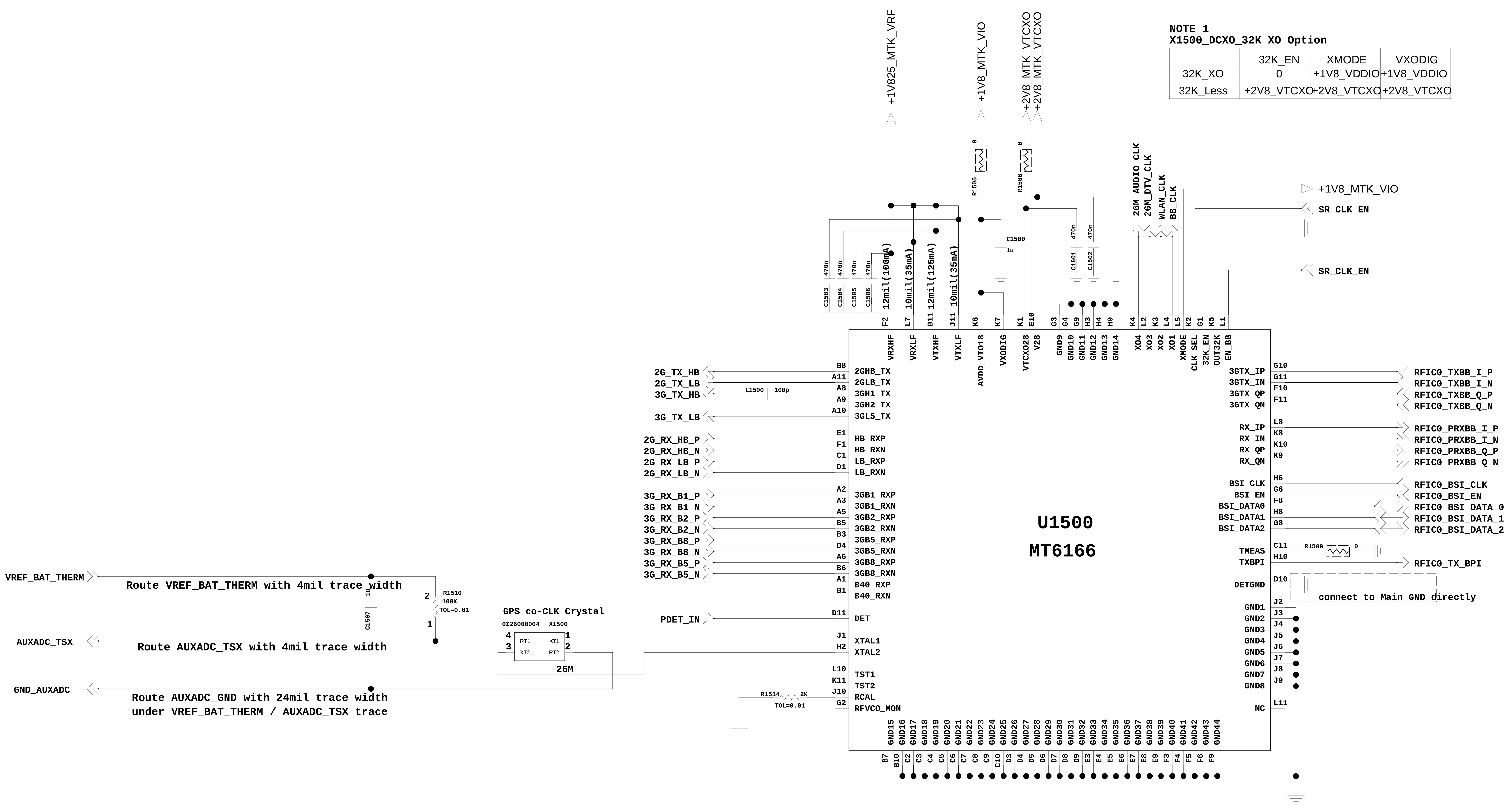
A2 PDM NUMBER Rev

User Name 1 1 Drawing Date

< 1-4-2-1_RFIC_MT6166 >   Rev_0.3

NOTE 1
X1500_DCXO_32K XO Option

	32K_EN	XMODE	VXODIG
32K_XO	0	+1V8_VDDIO	+1V8_VDDIO
32K_Less	+2V8_VTCXO	+2V8_VTCXO	+2V8_VTCXO



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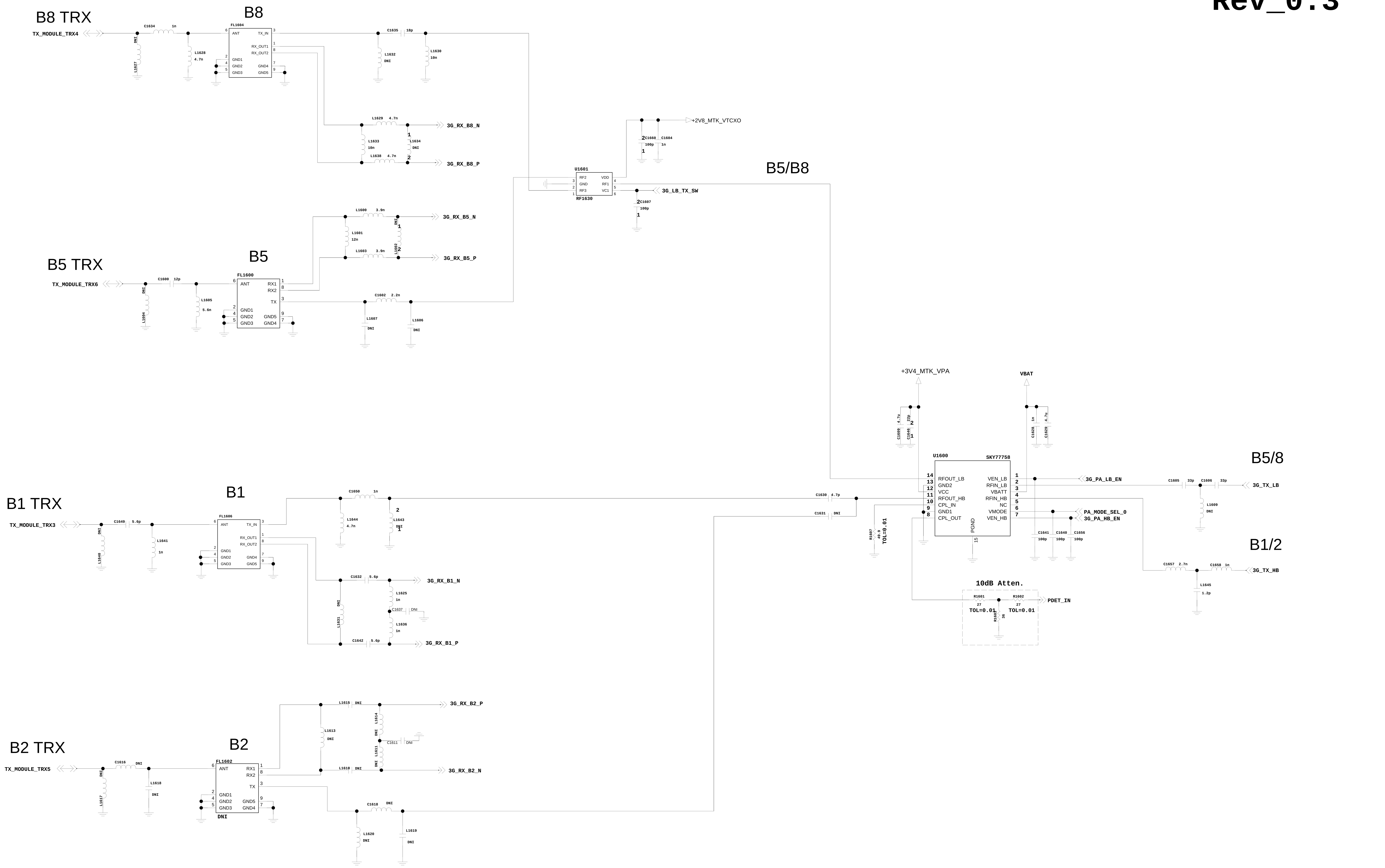
< 1-4-3-4_TRX_B1_B2_B5_B8_Quad band >

Rev_0.3



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Schematic1

A2 PDM NUMBER Rev

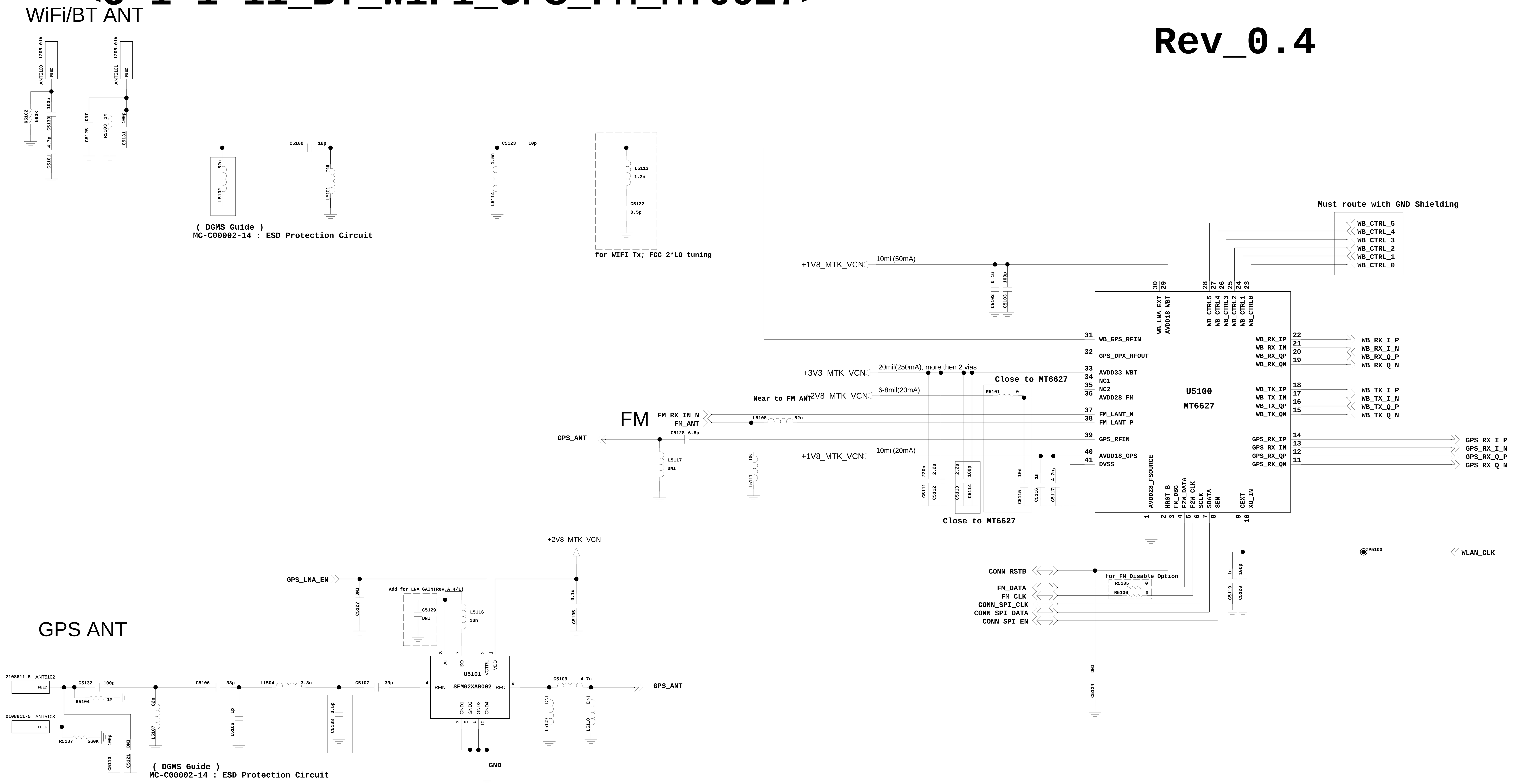
3 3

User Name

Drawing Date

<5-1-1-11_BT_WiFi_GPS_FM_MT6627>

Rev_0.4



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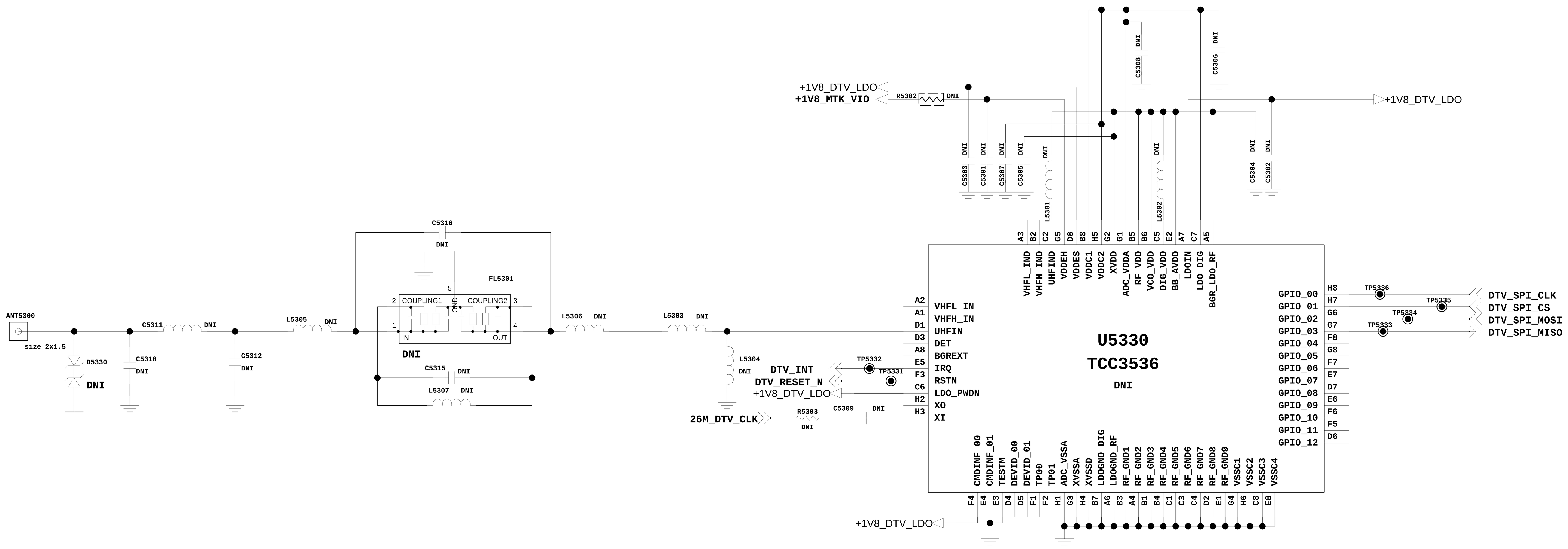
Schematic1

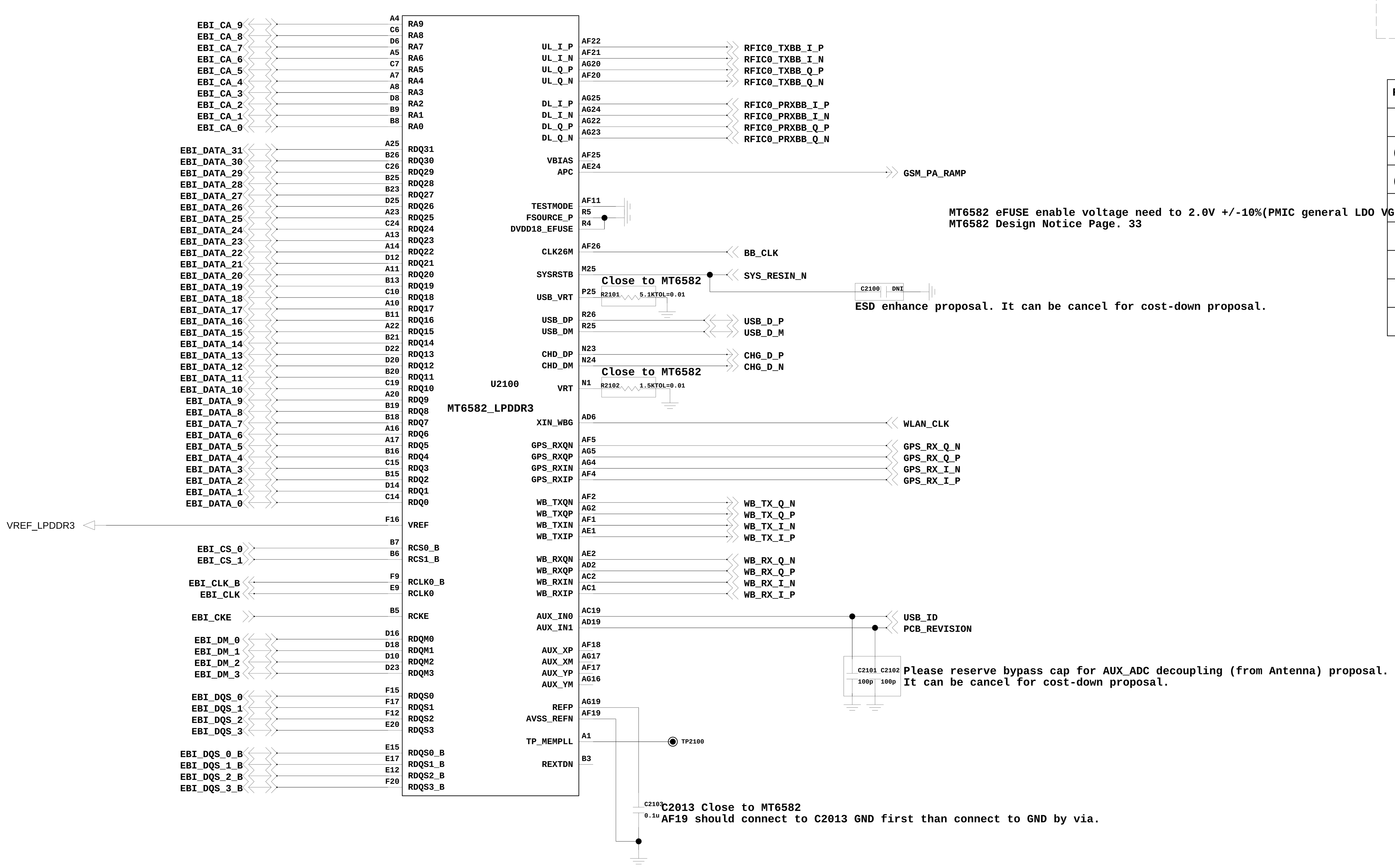
A2	PDM NUMBER	Rev	
4 of 4	Drawing Date	User Name	

<5-3-1-10_Full-SEG_TCC3536> Rev_1.0

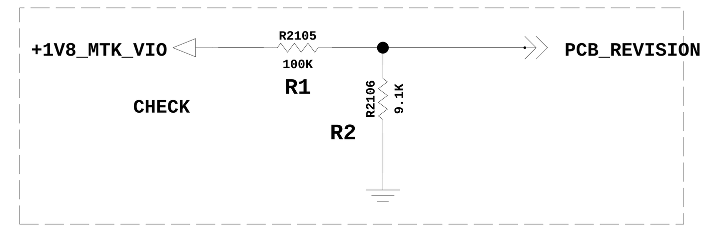


LG Electronics
Schematic1
A2 PDM NUMBER Rev
User Name
Drawing Date

<2-3-3-2-1_MT6582_DATA>



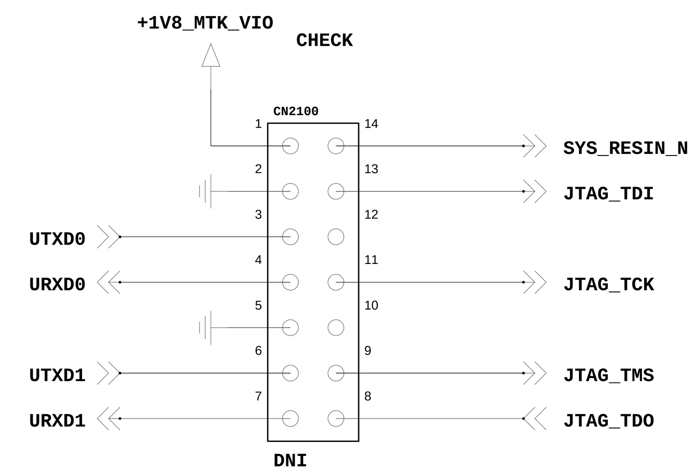
ADC



Revision	R1	R2	ADC Range(Voltage)
EVB1 (A)	100K	30K	29-51(0.415V)
EVB2 (TEST_B)	100K	51K	52-71(0.608V)
A (TEST_C)	100K	82K	72-91(0.811V)
B (NEW A)	100K	130K	92-111(1.017V)
C (NEW B)	100K	200K	112-128(1.200V)
1.0	100K	9.1K	8-28(0.150V)
1.1	100K	330K	130-150(1.381V)

< 9-3-1_JTAG >

Rev_0.4



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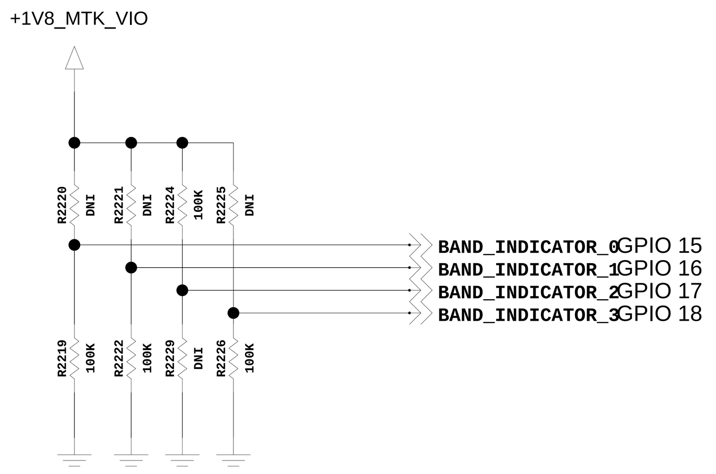
Schematic1

A2 PDM NUMBER Rev

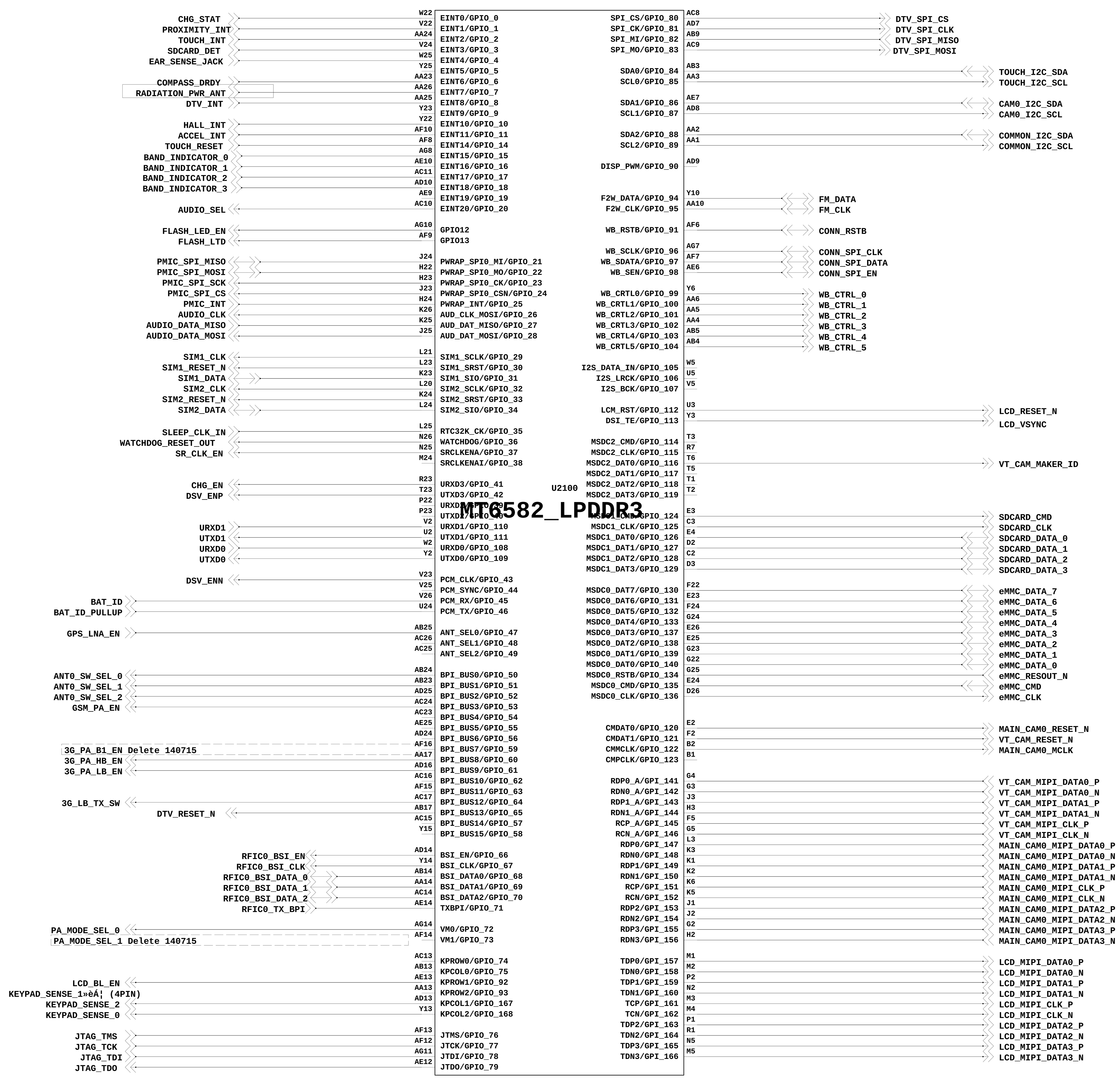
7 7 User Name Drawing Date

<2-3-3-2-2_MT6582_GPIO>

Model_Indicator



		GPIO 15	GPIO 16	GPIO 17	GPIO 18	SUM
H502TV	DUAL, 158	0	0	0	0	0
H502f	DUAL, 158	0	0	1	0	4
H502g	DUAL, 258	0	1	0	0	2
H500f	SINGLE, 158	1	0	0	0	1
H500TR	SINGLE, 158	1	0	1	0	5
H500g	SINGLE, 258	1	1	0	0	3

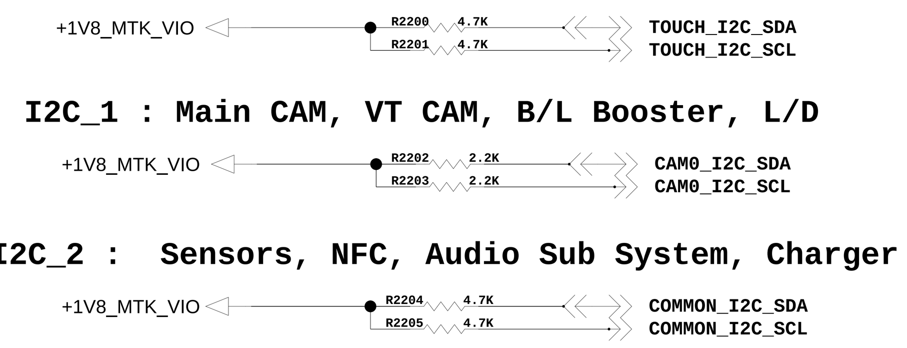


I2C Pull-Up

I2C_0 : Touch

I2C_1 : Main CAM, VT CAM, B/L Booster, L/D

I2C_2 : Sensors, NFC, Audio Sub System, Charger



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Schematic1

PDM NUMBER

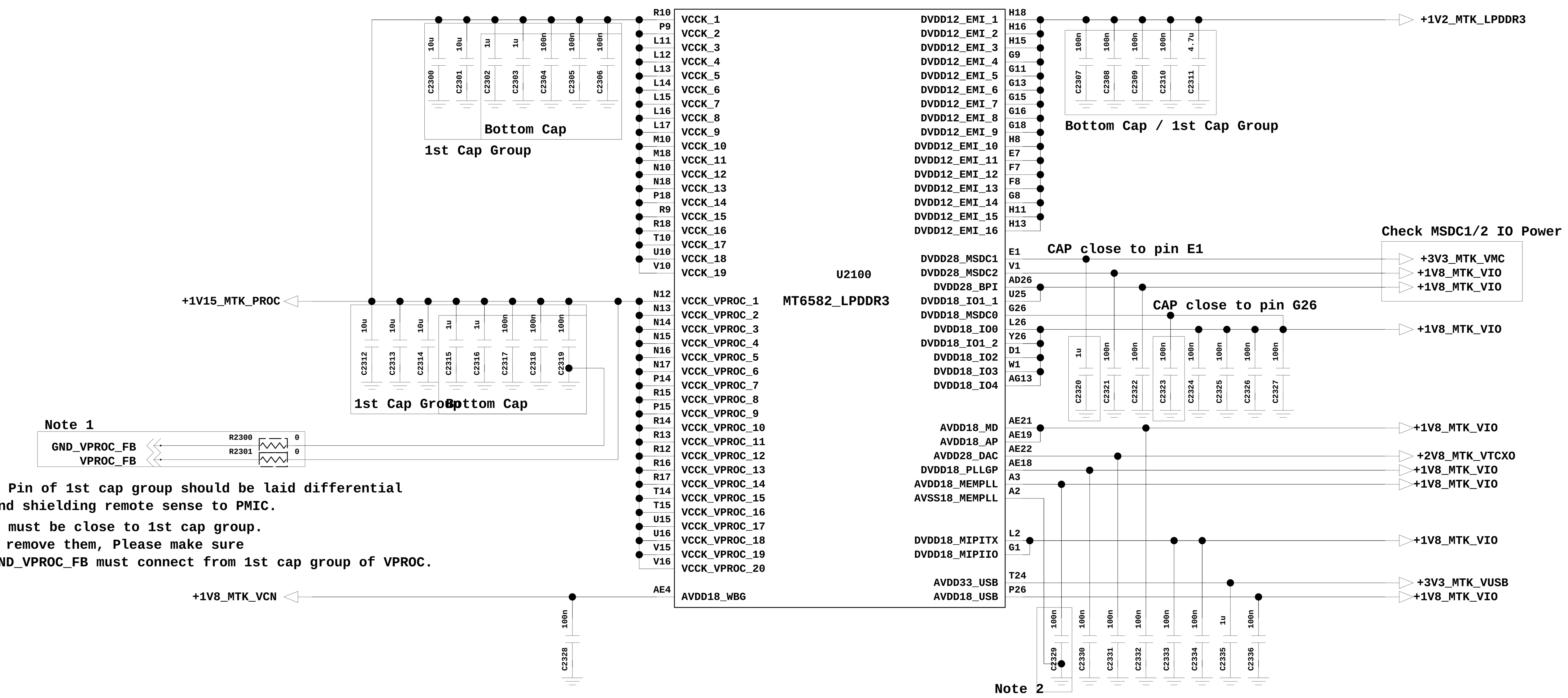
Rev

A2

User Name

Drawing Date

<2-3-3-2-3_MT6582_POWER>



Note 1
1. VPROC_BB, GND Pin of 1st cap group should be laid differential
pair with ground shielding remote sense to PMIC.
2. R2300 & R2301 must be close to 1st cap group.
If you want to remove them, Please make sure
the VPROC_FB/GND_VPROC_FB must connect from 1st cap group of VPROC.

Note 2 AVDD18_MEMPLL pin and AVSS18_MEMPLL pin should
be connected with PCB CAP first, and then connected
with PMU and PCB ground. CAP should be near MT6582
as possible as we can.

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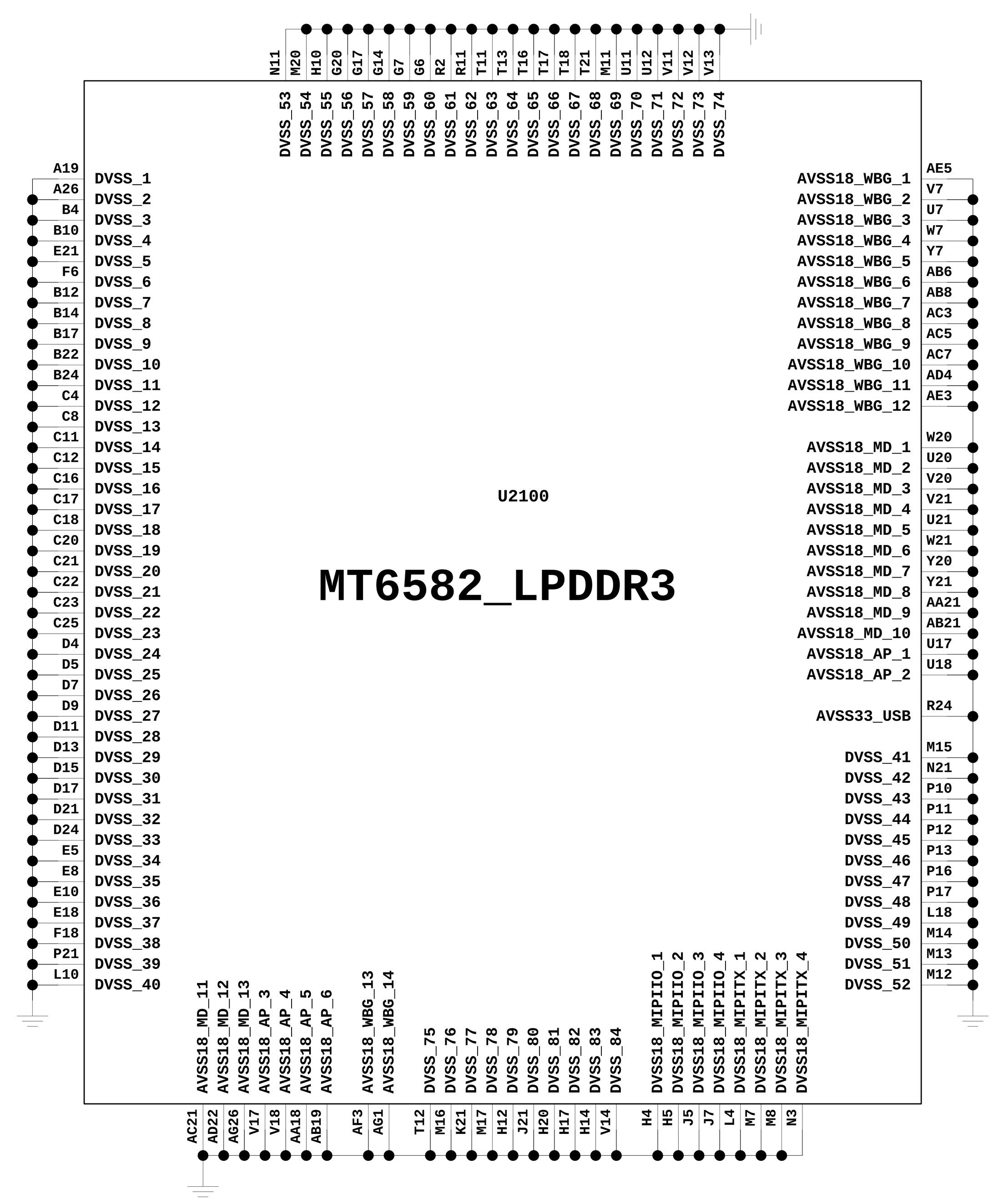
TITLE Schematic1

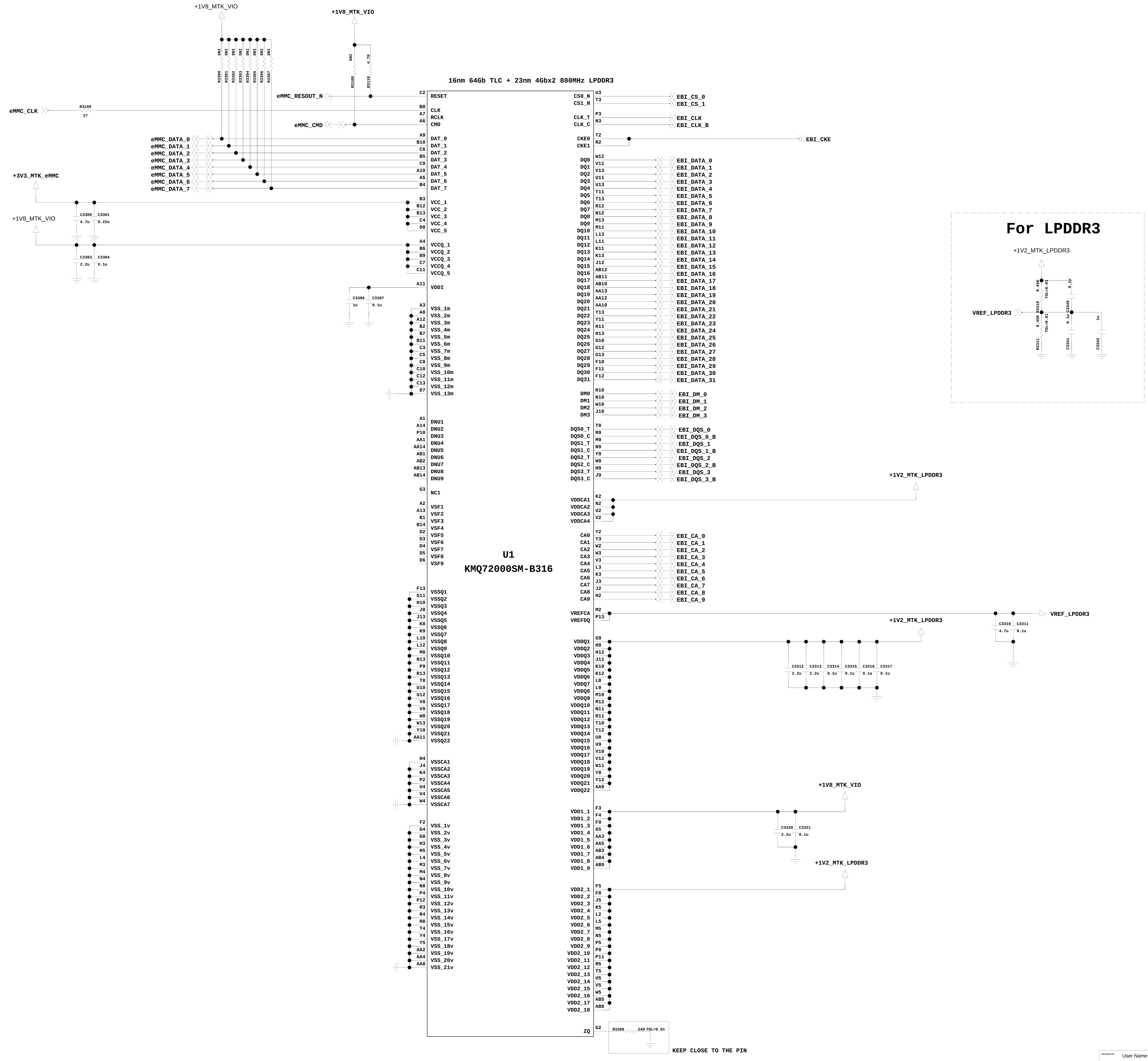
SIZE A2 PDM NUMBER REV Rev

User Name Drawing Date SHEET 9 of 9

<2-3-3-2-4_MT6582_GND>



< 3-3-3-2-1_MCP_eMMC_5_0_8G_DDR3_8G_Samsung > Rev 1.0



LG Electronics

Schematic1

A2 PDM NUMBER Rev

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User Name

Drawing Date

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<4-1-7-1_MT6323_DATA>
Rev_0.4
ISENSE/BSTSNS 4mil
differential to Rsense
VBAT
+4V9_VCHG_LDO
VBAT_THERM
Close to PMIC
CHG_D_N
CHG_D_P
WATCHDOG_RESET_OUT
PM_KYPD_PWR_N
SYS_RESIN_N
PMIC_INT
SR_CLK_EN
PMIC_SPI_SCK
PMIC_SPI_CS
PMIC_SPI_MOSI
PMIC_SPI_MISO
Note 1
Cap must to be close
to PMIC AUXADC_AUXIN_GPS pin
VREF_BAT_THERM
AUXADC_TSX
Cap must to be close to PMIC
GND_AUXADC
HOOK_DETECT
Don't place any capacitor on 26MHz clock
between MT6166 & MT6323
26M_AUDIO_CLK
SLEEP_CLK_IN
32.768KHz
Note 2
Connect TSX/XTAL GND
to AUXADC_GND first
than connect to main GND.
Refer to GPS co-clock layout rule
U4100
MT6323
BATSNS
ISENSE
BATON
VCDT
VDRV
CHRLDO
CHG_DM
CHG_DP
PWRKEY
SYSRSTB
RESETB
FSOURCE
INT
EXT_PMIC_EN
PMU_TESTMODE
SRCLKEN
FCHR_ENB
SPI_CLK
SPI_CSN
SPI_MOSI
SPI_MISO
AUXADC_VREF18
AUXADC_AUXIN_GPS
AVSS28_AUXADC
ACCDET
CLK26M
RTC_32K1V8
RTC_32K2V8
XIN
XOUT
SIM1_AP_SCLK
SIMLS1_AP_SIO
SIM1_AP_SRST
SIM2_AP_SCLK
SIMLS2_AP_SIO
SIM2_AP_SRST
SIMLS1_SCLK
SIMLS1_SIO
SIMLS1_SRST
SIMLS2_SCLK
SIMLS2_SIO
SIMLS2_SRST
VBAT_SPK
GND_SPK
AU_MICBIAS0
AU_MICBIAS1
AU_VIN0_P
AU_VIN0_N
AU_VIN1_P
AU_VIN1_N
AU_VIN2_P
AU_VIN2_N
SPK_P
SPK_N
AU_HSP
AU_HSN
AU_HPL
AU_HPR
AUD_MOSI
AUD_CLK
AUD_MISO
SIM1_CLK
SIM1_DATA
SIM1_RESET_N
SIM2_CLK
SIM2_DATA
SIM2_RESET_N
SIM1_CLK_PM
SIM1_DATA_PM
SIM1_RESET_PM
SIM2_CLK_PM
SIM2_DATA_PM
SIM2_RESET_PM
Note 3
If you use digital MIC, please change cap to 1.0uF
MIC_BIAS_1
MIC_BIAS_2
MIC_IN1_P
MIC_IN1_N
EAR_MIC_JACK
MIC_IN2_N
MIC_IN3_P
MIC_IN3_N
RCV_P
RCV_N
HPH_L
HPH_R
AUDIO_DATA_MOSI
AUDIO_CLK
AUDIO_DATA_MISO
<9-1_Battery_CNT_4Pin>
Rev_1.1
Routing Width 2mm
If battery NTC is 10kohm, R9532=16.9K, R9535=27K
If battery NTC is 47kohm, R9532=61.9K, R9535=100K
BAT_ID
BAT_ID_PULLUP
CN9100
R9532, R9535 must to be close to PMIC AUXADC_REF pin
(DGMS Guide)
MC-C00073-15 : Apply to large capacity around the bat connector
1. Over 47uF around the bat connector
2. Over 100uF totally on Vbat line
Use the TVS diode of VR=5V, when TVS diode have to use for ESD
(DGMS Guide)
MC-C06162-1 : 1. Do not connect negative of tantal Cap. to positive voltage if use
2. We recommend Ceramic Cap.
(DGMS Guide)
MC-C02048-18 : 4. Polymer type tantal cap. is Spec =< Voltage rating X 0.8
<9-2-1_USB_IO>
Rev_1.1
VBUS_USB_IN
CN10001
USB_D_M
USB_D_P
+1V8_MTK_VIO
USB_ID
<4-7-1-1_Linear Charger>
Rev_0.3
+4V9_VCHG_LDO
+1V8_MTK_VIO
U4700
RT9536GQW
PGB Pin Connect PM8x26 for Boot System
Wireless charger Interrupt
CHG_STAT
CHG_EN
NOTE1
NOTE2
NOTE1
ISET Setting
560 ohm : 910mA
680 ohm : 820mA
750 ohm : 750mA
820 ohm : 680mA
NOTE2
IEOC Setting
3.3K ohm 16.5%
2K ohm 10%
LG Electronics
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PDM NUMBER
User Name
Drawing Date
LGE Internal Use Only

<4-1-7-2_MT6323_POWER> REV_0.4

Note 1
Refer to MT6323 design notice for Buck GND layout rule.
Choise input cap MT6572 between MT6582

Ref.	MT6572	MT6582
C4136	DNI	1uF
C4137	10uF	4.7uF
C4138	4.7uF	4.7uF
C4139	10uF	4.7uF
C4150	DNI	4.7uF

Buck converter brief specifications

VSYS	0.7-1.3V/1.35V (optional) (6.25mV/step)	2,800	Processor / Digital Core
VSYS	2.2	1,400	System LDO input
VPA	0.5V - 3.4V (0.1V/step)	600	3G power amplifier

LDO types and brief specifications

Type	LDOname	Vout (Volt)	Imax (mA)	Application
ALDO	VCN28	2.8	30	RF chip
ALDO	VTCXO	2.8	40	13/26MHz reference clock
ALDO	VA	2.8	150	Analog baseband
ALDO	VCAMA	2.8	150	Analog power for camera module
DLDO	VCN_3V3	3.3/3.4/3.5/3.6	240	WIFI
DLDO	VIO28	2.8	200	Digital IO
DLDO	VSIM1	1.8/3.0	50	1st Sim card
DLDO	VSIM2	1.8/3.0	50	2st SIM card
DLDO	VSUB	3.3	20	USB
DLDO	VGP1	1.2/1.3/1.5/1.8 2.0/2.8/3.0/3.3	100	General purpose LDO
DLDO	VGP2	1.2/1.3/1.5/1.8 2.0/2.5/2.8/3.0	100	General purpose LDO
DLDO	VEMC_3V3	3.0/3.3	400	3.3V EMMC
DLDO	VCAM_AF	1.2/1.3/1.5/1.8 2.0/2.8/3.0/3.3	100	AF application
ALDO	VMC	1.8/3.3	100	SD 2.0/3.0 memory card
ALDO	VMCH	3.0/3.3	400	SD 3.0 memory card
DLDO	VIBR	1.2/1.3/1.5/1.8 2.0/2.8/3.0/3.3	100	Vibrator
PTCLDO	VRTC	2.8	2	Real-time clock
VSYS LDO	VM	1.24/1.39/1.54/1.84	700	Memory power
VSYS LDO	VRF18	1.825	200	RF application
VSYS LDO	VIO18	1.8	300	IO pad power
VSYS LDO	VCAMD	1.2/1.3/1.5/1.8	150	Camera application
VSYS LDO	VCAM_IO	1.8	100	Camera IO application
VSYS LDO	VGP3	1.2/1.3/1.5/1.8	200	General purpose LDO
VSYS LDO	VCN18	1.8	120	General purpose LDO

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A2 PDM NUMBER Rev
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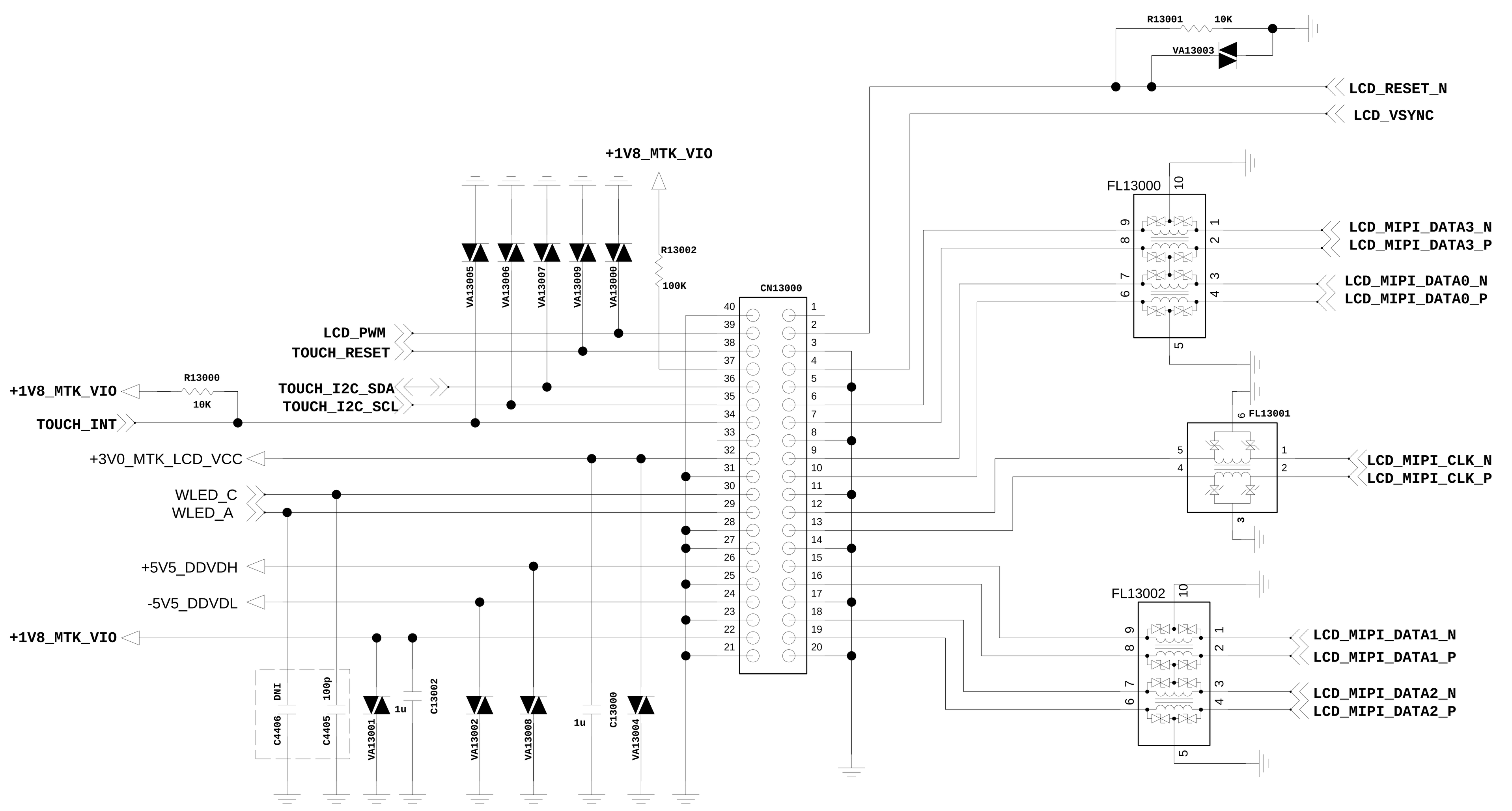
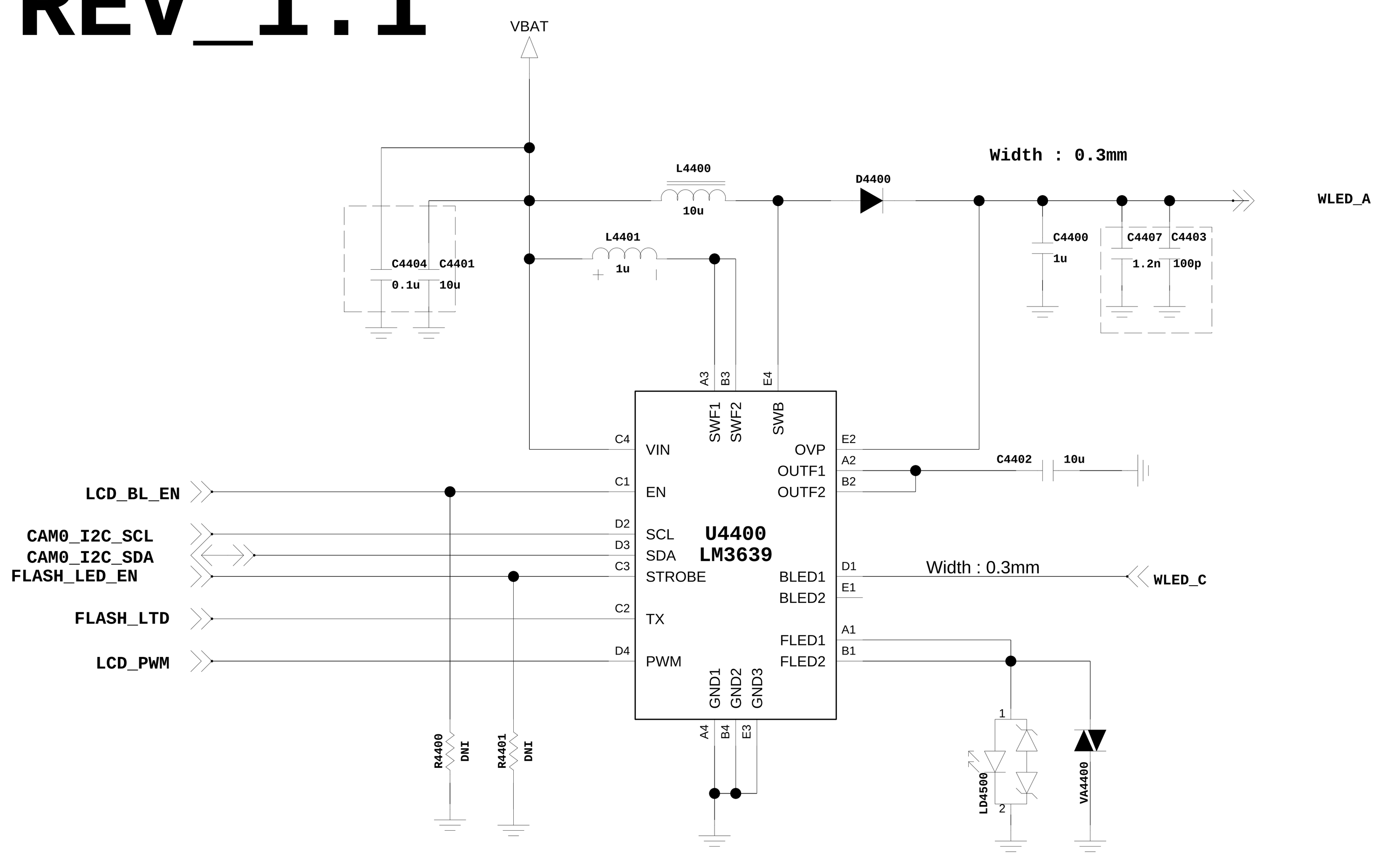
User Name Drawing Date

<7-1-5_LCD_HD_V1_1>

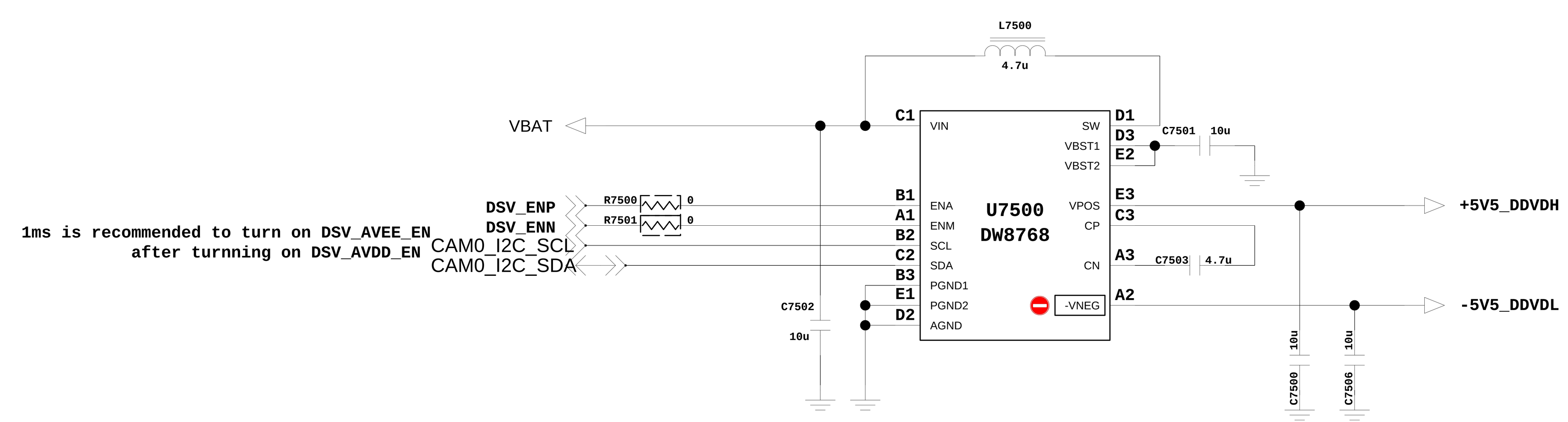


<4-4-6_LCD_Backlight_Flash_IC_LM3639>

REV_1.1



<7-5-8_DSV_DW8768> Rev_0.3



DGMS Guide
MC-C06162-1 : 1. Don't connect Positive of Tantal, TVS diode to Negative voltage if use
2. We recommend MLCC capacitor in this case

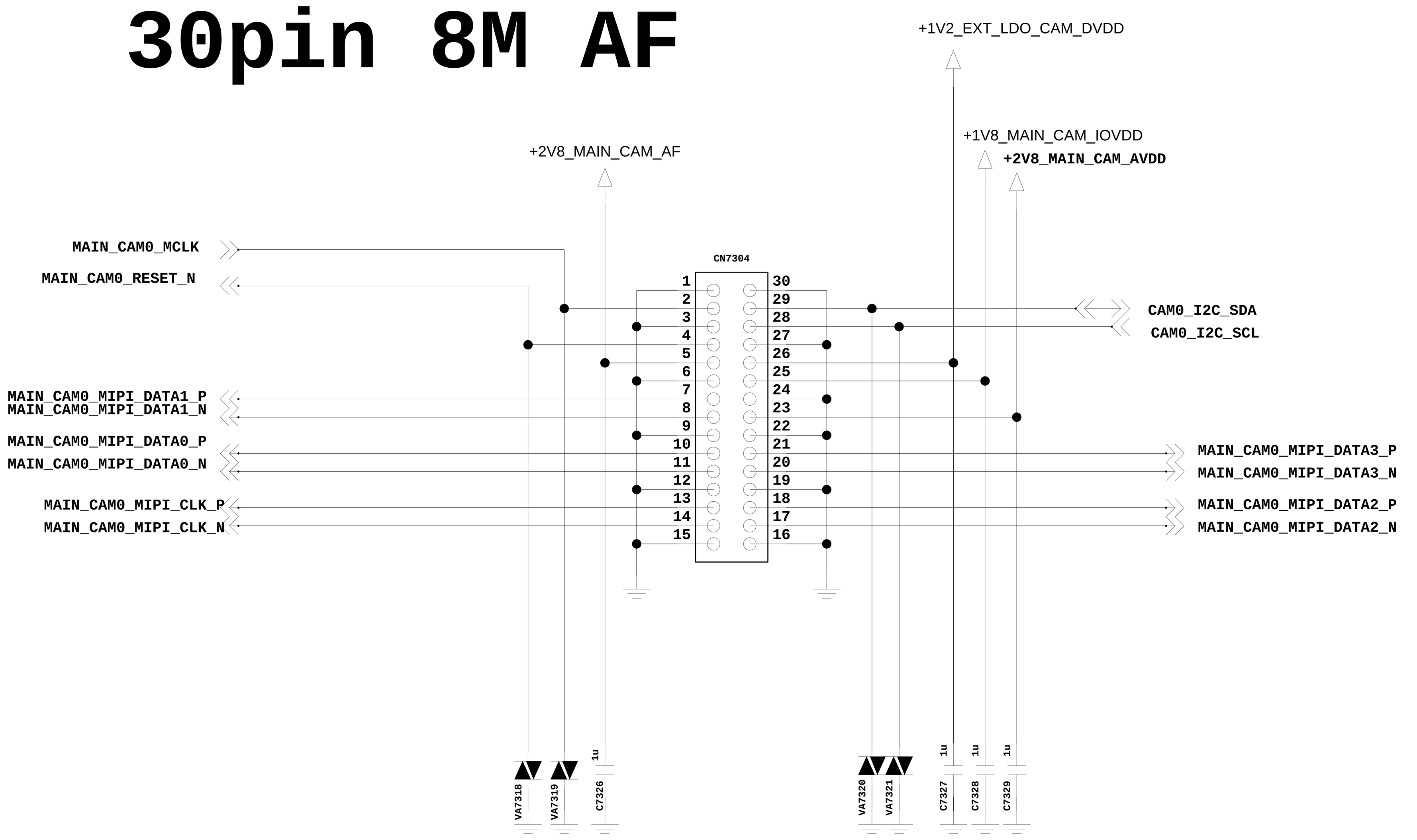
LG Electronics

Schematic1

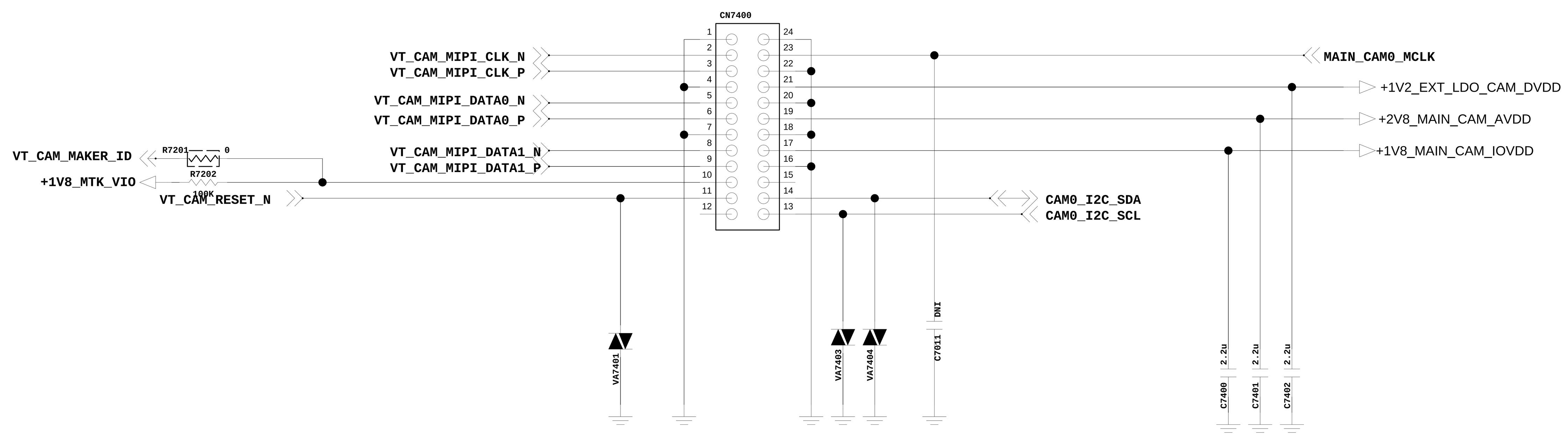
SIZE	DWG NO		REV
A2	PDM NUMBER		Rev
SHEET 14 of 14	User Name	Drawing Date	

< 7-3_Main_Camera > Rev_1.1

30pin 8M AF



< 7-4_VT_Camera> Rev_1.1



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A2 PDM NUMBER Rev

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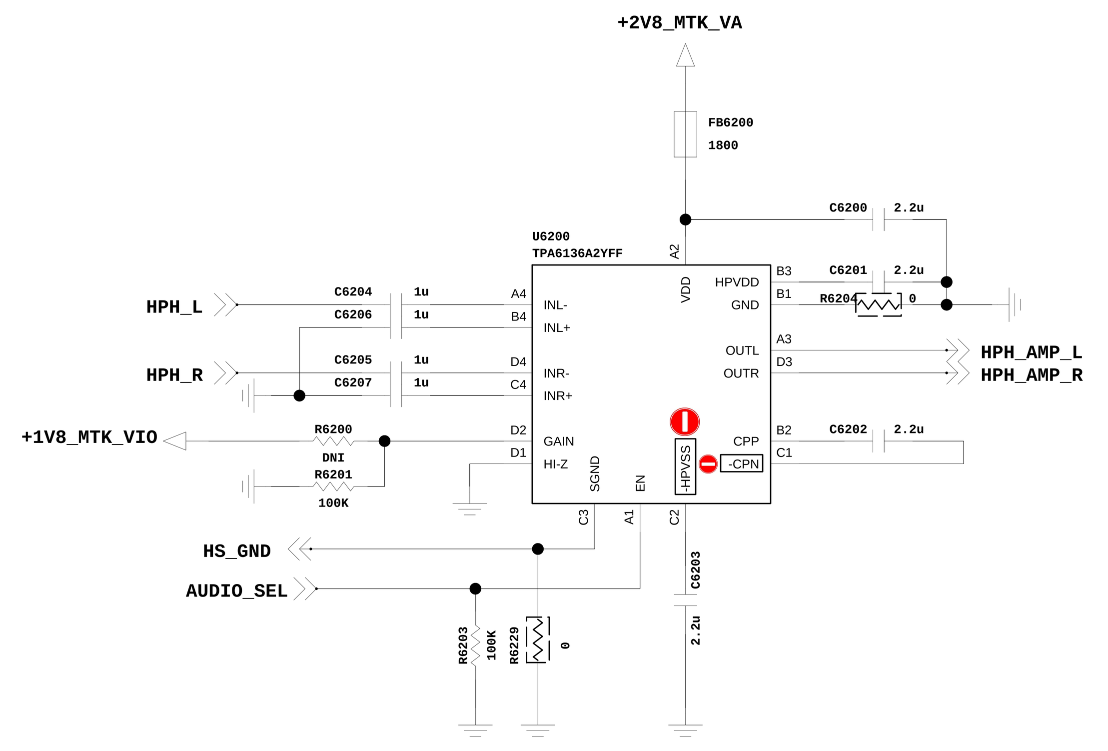
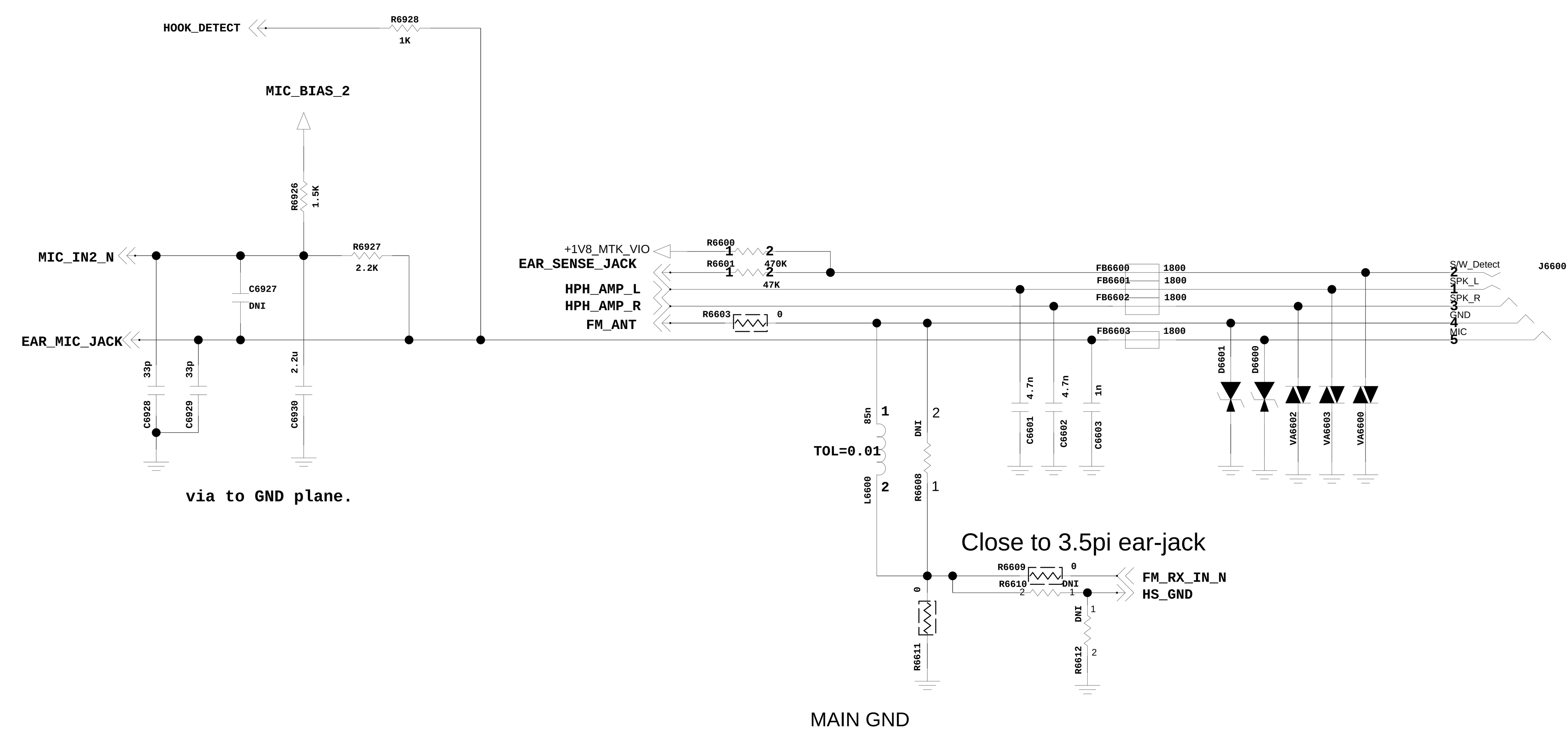
<6-2-1-1_Headphone_AMP_TAP6136>

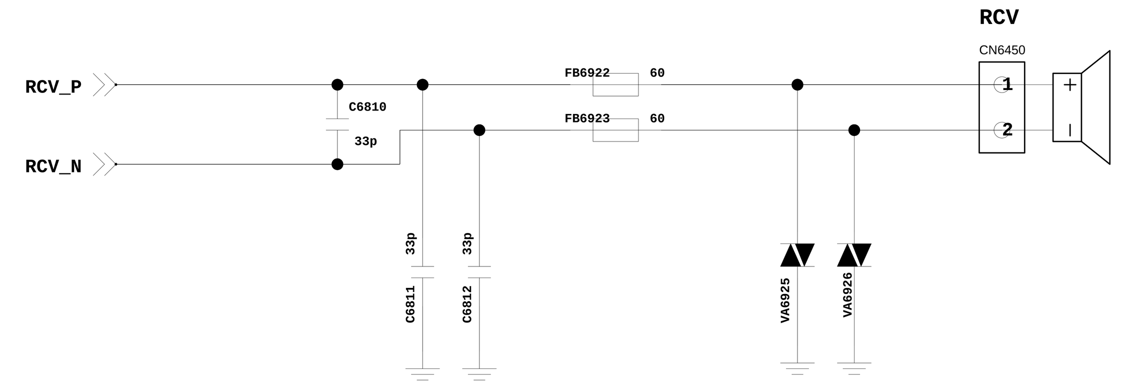


< 6-6-1_Earjack >



< 6-8-2_Receiver >

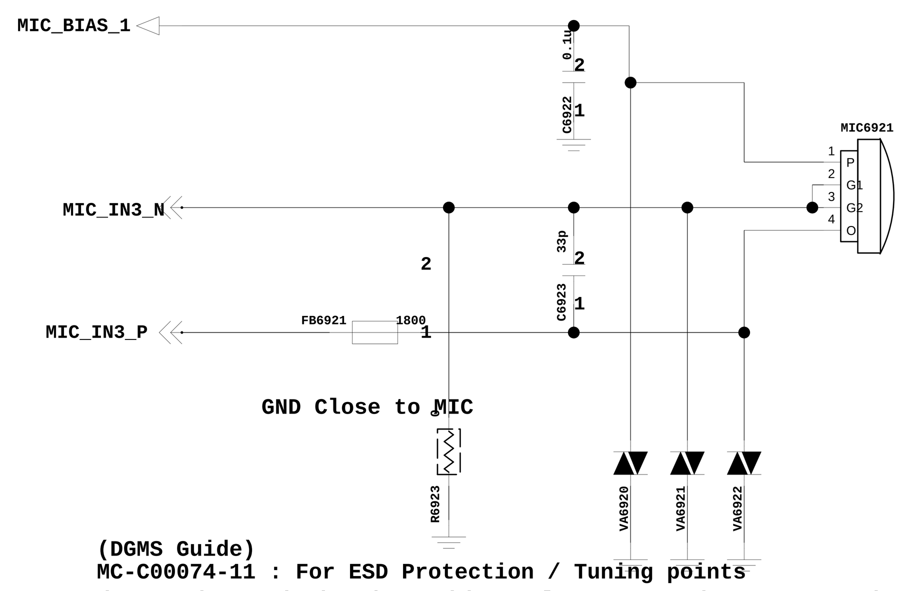
Rev_0.4



(DGMS Guide)
MC-C04773-8 : For ESD Protection / Tuning points
When USE) TVS devices's Working Voltage must be greater than the Path voltage

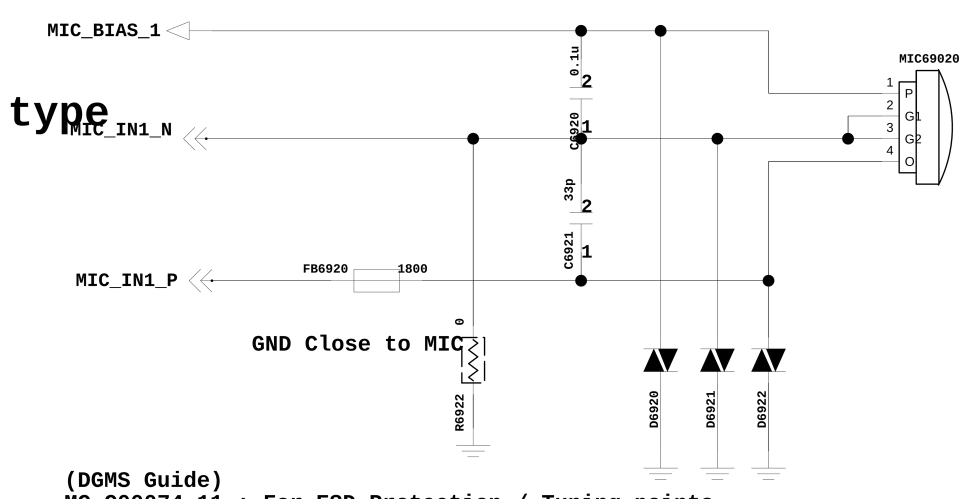
< 6-9-2_2MIC > Rev_0.4

SUB_MIC



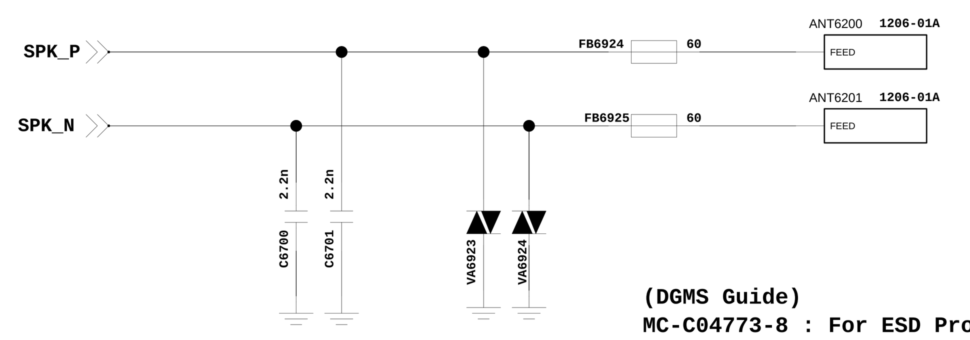
(DGMS Guide)
MC-C00074-11 : For ESD Protection / Tuning points
When USE) TVS devices's Working Voltage must be greater than the Path voltage

MAIN_MIC



(DGMS Guide)
MC-C00074-11 : For ESD Protection / Tuning points
When USE) TVS devices's Working Voltage must be greater than the Path voltage

<6-7-1_Speaker> Rev.1.0



(DGMS Guide)
MC-C04773-8 : For ESD Protection / Tuning points
When USE) TVS devices's Working Voltage must be greater than the Path voltage
should apply capacitor under 50pF value for ESD protection with d-class amp

LG Electronics
TITLE Schematic1
A2 PDM NUMBER Rev
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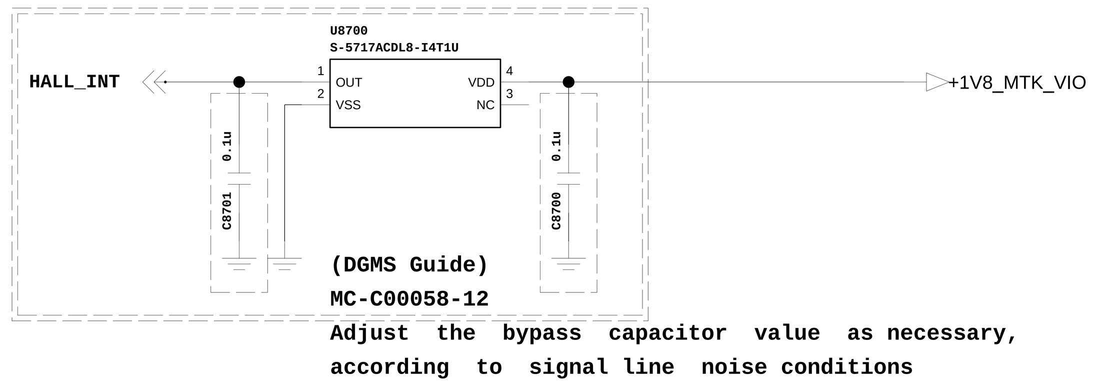
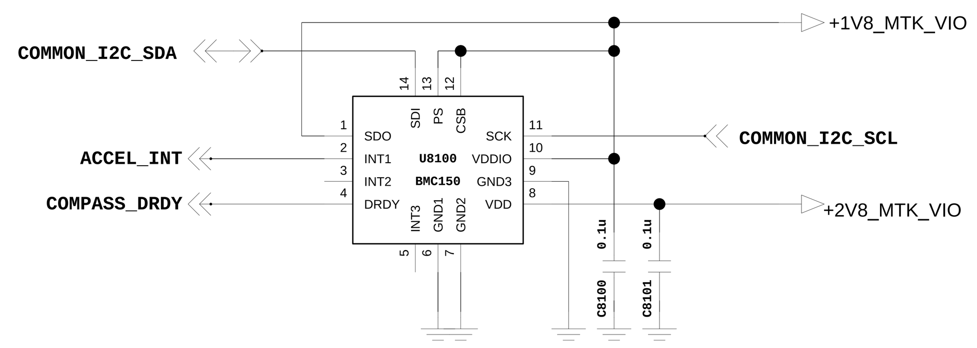
User Name Drawing Date

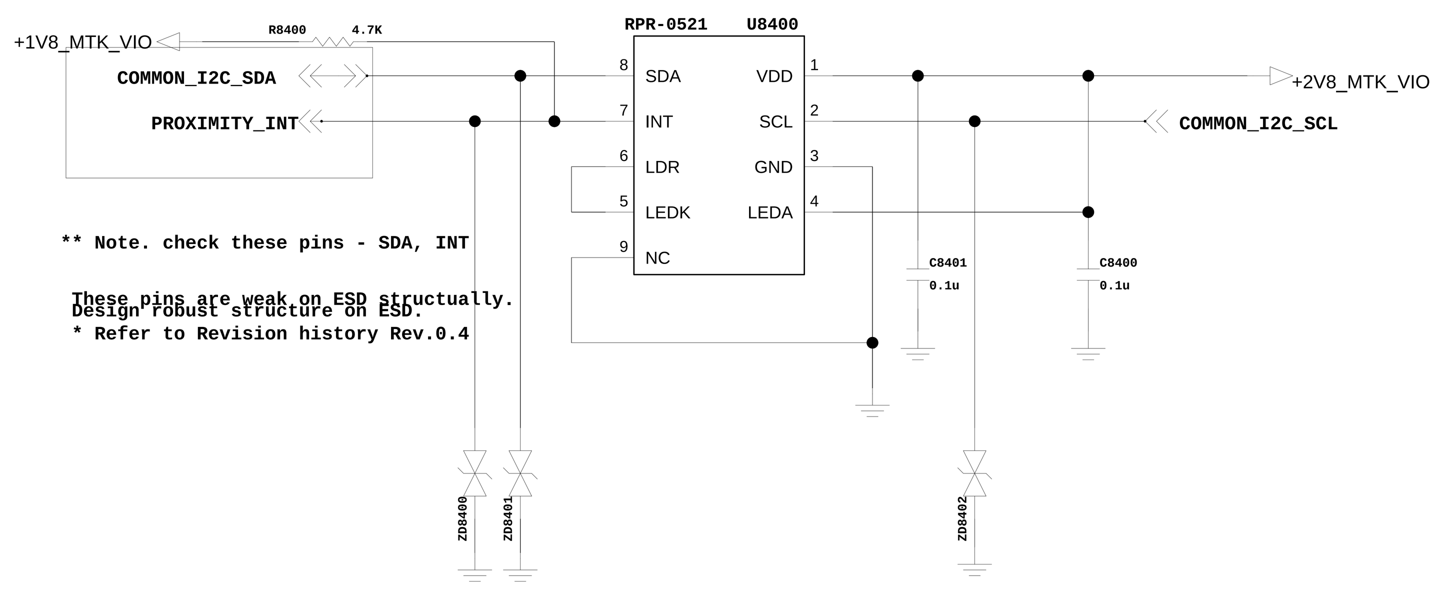
<8-7-1-5_Hall_IC_S5717> Rev_1.1



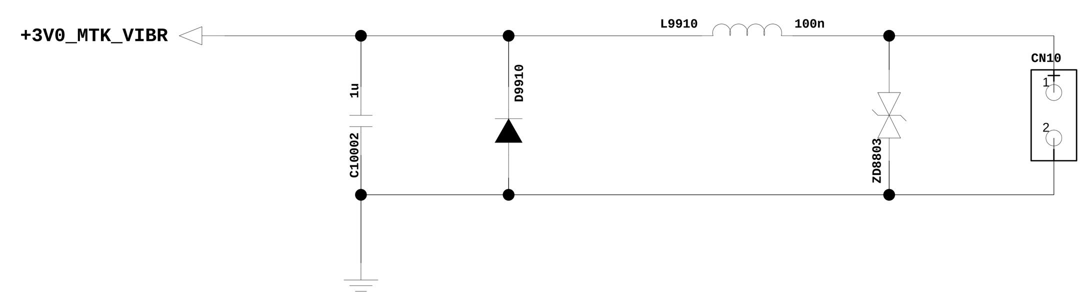
<8-1-2-1_Accel_Compass_BMC150> REV_1.0



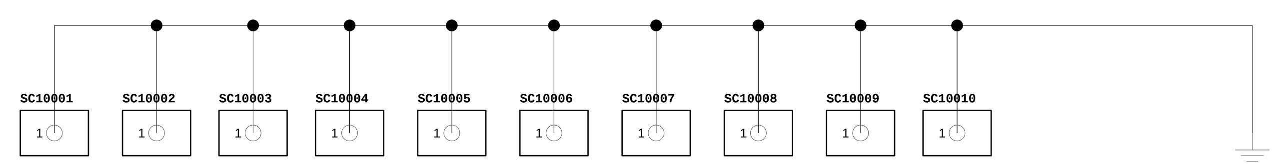
<8-4-3-5_Proximity_RPR-0521> Rev_0.3



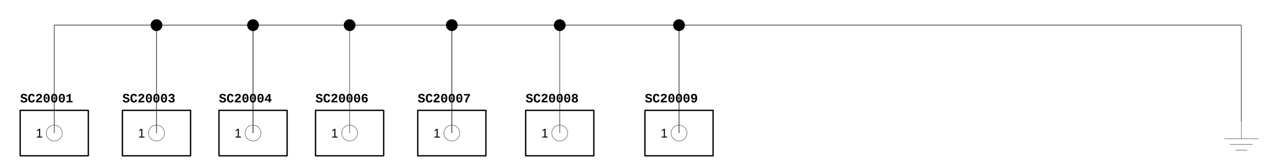
< 9-9-1-1_Motor > Rev_0.3



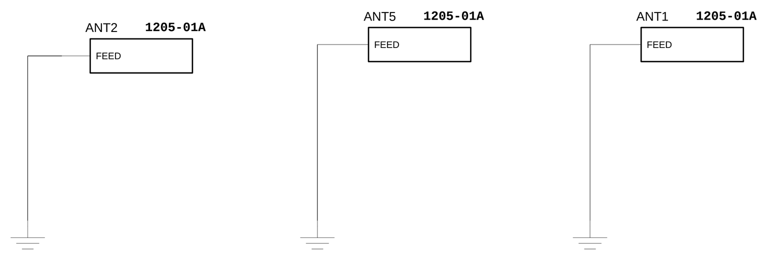
TOP SHIELD CAN CLIP



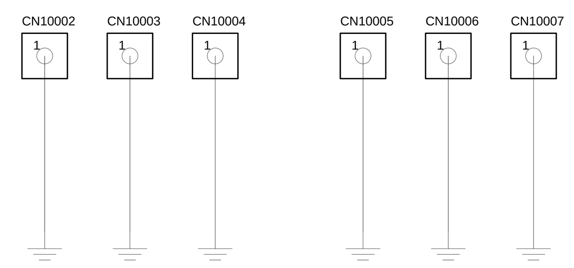
BOTTOM SHIELD CAN CLIP



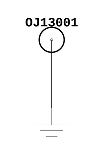
PCB Contact



SMT GASKET



SCREW_HOLE



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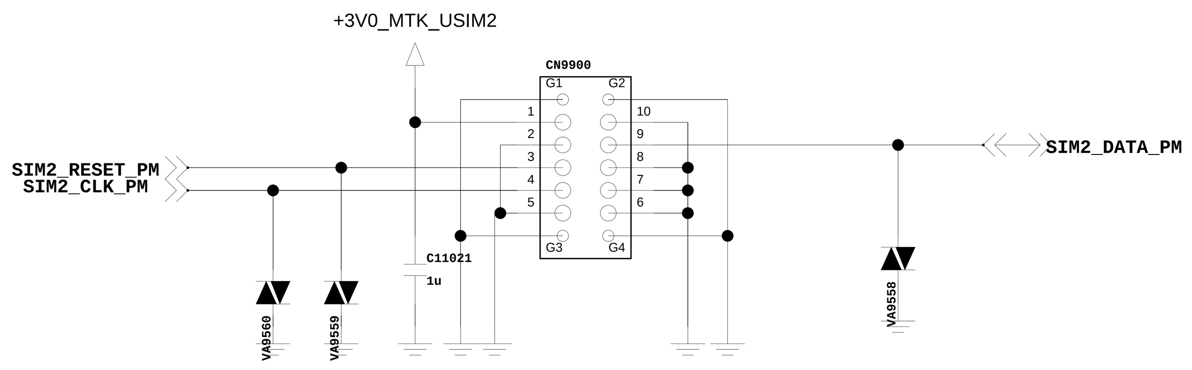
Sub SIM Connector



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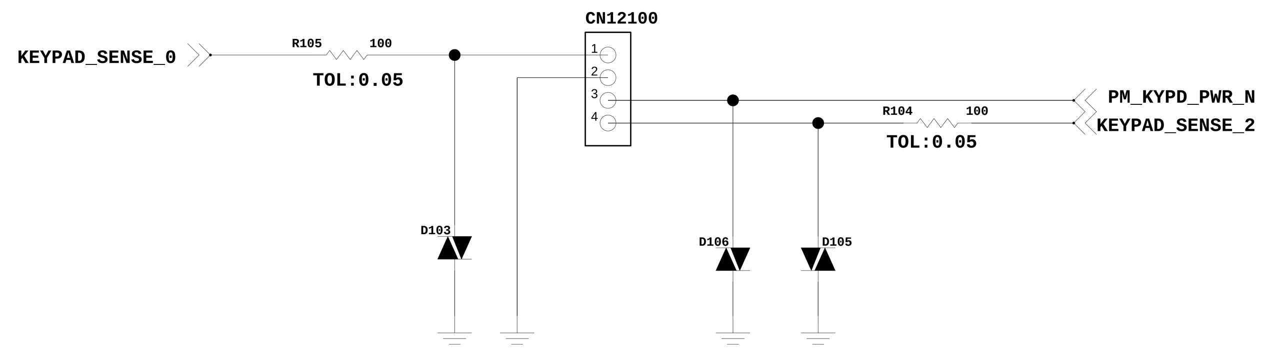

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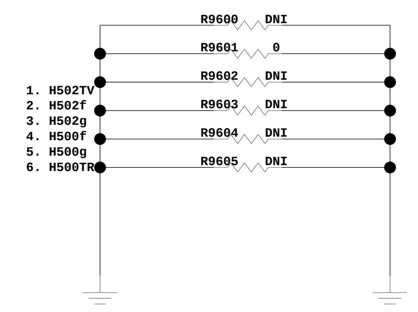

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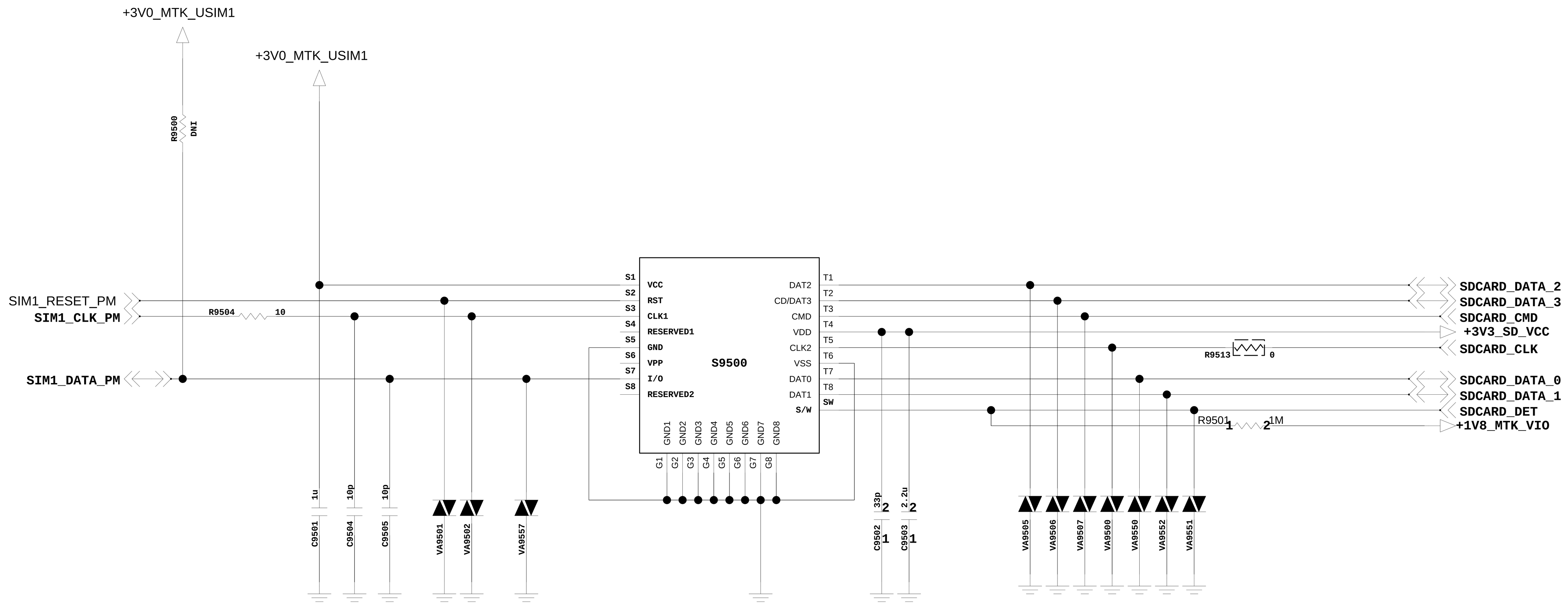

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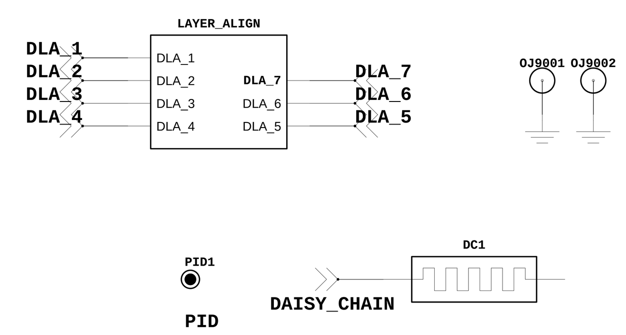

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Rev	
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User Name	Drawing Date

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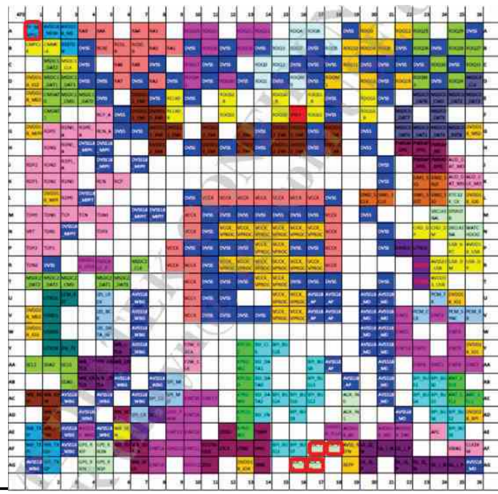
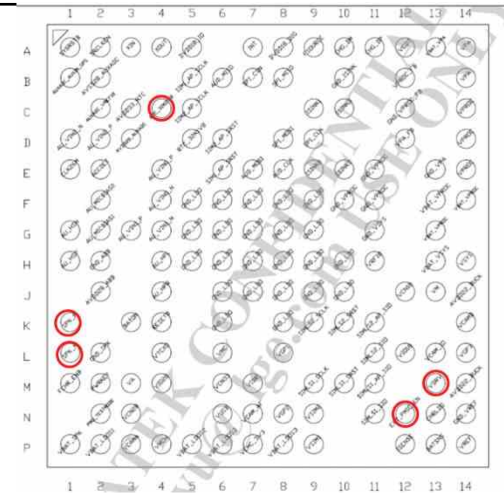
 O Not used	 O Not used
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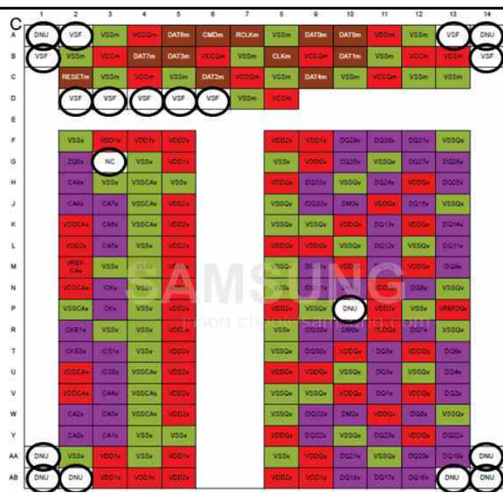 O Not used	